

A predictive coding model for memory replay in the neocortex

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Abstract. The neocortex and the hippocampus are two complementary learning systems which interact during memory construction and consolidation. The hippocampus stores episodic memories coming from the neocortex, and later replays them back to the neocortex to transform them into semantic memory during memory consolidation. It was found that the hippocampus is also involved in imagining novel, never experienced before episodes, suggesting the idea that hippocampal replay is a generative process. We propose a predictive coding model of the neocortex and model its interactions with the hippocampus during memory replay. Our work suggests that the quality of replay depends on the separability of the representations, which can be controlled by the width and depth of the network.

Keywords: Predictive coding · Memory · Representation learning.

1 Introduction

According to McClelland and McNaughton [13], the hippocampus and the neocortex are complementary learning systems (CLS). The neocortex is responsible for semantic memory, that is the general knowledge that we have about the world, whereas hippocampus stores episodic memories, which correspond to an individual's emotional and sensory experiences, [11]. After storage in the hippocampus, an episodic memory can be recalled from a corrupted version of it. Furthermore, episodic memories are replayed during rest or sleep for memory consolidation. Memory replay corresponds to the spontaneous reactivation of the activity associated to an episodic memory in the hippocampus and its subsequent reinstantiation in the neocortex.

According to the CLS theory, the reinstatement of new memories interleaves them with existing memories to integrate them into the semantic memory of the neocortex. This principle has inspired successful applications in reinforcement learning and supervised continual learning tasks, where the learning of new tasks interferes with the knowledge of previously learned tasks [14,5]. Subsequently, experimental evidence suggested that the hippocampus is engaged in

mental simulation and imagination, beyond episodic memory [25,20]. This has inspired researchers from NeuroAI to implement hippocampal replay as a generative process, in the form of “generative replay”, which exhibits some advantages over experience replay [22,24].

Despite the importance of memory replay for continual learning, its implementation in the neocortex is not well understood. Because of the similarity of architecture in all its areas, researchers hypothesized that a common algorithm underlies computations in the neocortex [8,9]. Principles of organization in the neocortex have emerged from empirical studies in the visual cortex, which have shown that this region is arranged hierarchically, with forward connections from lower to higher areas, and backward connections from higher to lower areas [8]. Predictive coding networks (PCN), originally proposed by Rao and Ballard [18], were later shown by Friston [8] to align closely with the neocortex’s architecture and information processing dynamics, establishing predictive coding as a compelling framework for understanding cortical computation. Since the foundational work of Friston [8], numerous computational models have demonstrated that PCN can effectively account for a wide range of functional properties observed along the visual cortical hierarchy [3,6,19,12].

Located at the apex of the neocortical hierarchy [2], the hippocampus receives input from the entorhinal cortex, which combines representations from different high-level neocortical areas of different sensory modalities, through forward connections (Fig. 1a). Thus, Bogacz and colleagues [19,23] modelled the hippocampus as the top level of a predictive coding network. However, by modelling the hippocampus and the neocortex as auto-associative memory systems within the same predictive coding framework, they overlooked the complementary roles these structures play in learning. In a previous study [7], we re-evaluated the role of the neocortex in auto-associative memory. We reproduced the result of Salvatori et al. [19] that PCNs can store training images as memories (Fig. 1c), but showed that this is done by overfitting an overparameterized network to a few training images. This work showed that while the neocortex might not be able to store patterns like the hippocampus, it integrates episodic memory in a semantic form during memory replay.

Memory replay can be decomposed into two phases, which we call hippocampal replay and cortical replay respectively (Fig. 1b). Memory replay is initiated in the hippocampus by hippocampal replay, which generates an episode in the entorhinal cortex. Then, cortical replay transfers the episode from the entorhinal cortex to the different neocortical areas through backward connections. In this paper, we propose to model the two phases of memory replay using a 2-layer PCN, focusing on the visual pathway. Experience and generative replay are two modes of hippocampal replay. Experience replay corresponds to the reactivation of a stored pattern of activations in the hippocampus, whereas generative replay activates a novel pattern of activations in the hippocampus. In both cases, the pattern of activations in the hippocampus is transferred to the neocortex during the subsequent cortical replay phase. Whereas hippocampal replay is modelled at a coarse, computational level, we model cortical replay at a more fine grained,

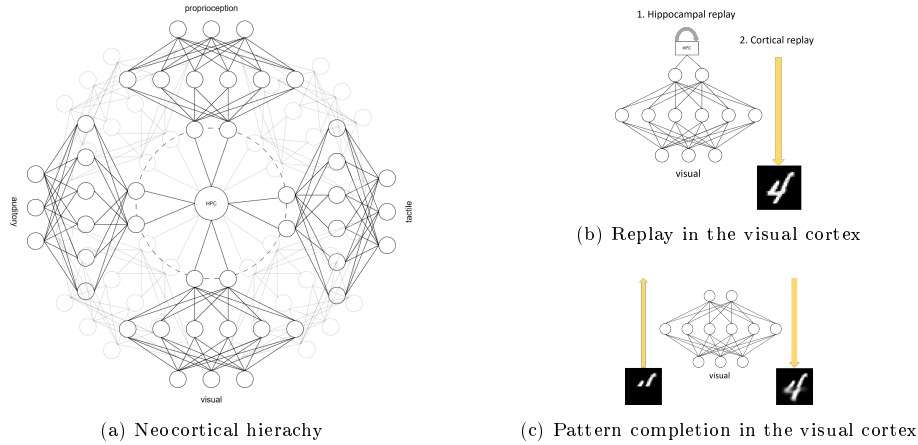


Fig. 1: (a) The high-level representations in neocortical areas corresponding to different sensory modalities are combined in the entorhinal cortex (represented as the dotted circle) which is the input of the hippocampus (HPC), located at the apex of the neocortical hierarchy. (b) After storage of an episodic memory in the hippocampus, it can be replayed in the hippocampus (hippocampal replay) before being reinstated in the neocortex, including the visual cortex (cortical replay). (c) The visual cortex can complete an image based on semantic memory [7].

algorithmic level. Our experiments study the effect of the depth and width of the PCN on the representations and on replay. Results are presented for the MNIST dataset, and discussed in light of additional experiments using Fashion-MNIST and CIFAR-10.

2 Related works

Models of memory replay. Replay in the neocortex was modelled computationally to study memory consolidation, using artificial neural network, such as Boltzmann machines [10] or Variational Autoencoders [21], which are not faithful to the architecture and information processing of the neocortex.

Image generation in PCN. Predictive coding networks (PCN) were used for the generation of inputs, but without specifying how generative processing in the neocortex can coexist with the presence of sensory inputs, or its relation to the hippocampus. On the one hand, Monte Carlo predictive coding [15] was proposed for learning probability distributions of sensory inputs, arguing that classical PCN demonstrated limited performance in generative tasks. On the other hand, Ororbia and Kifer [17] generated images from classical PCNs, but only a few samples were shown, leaving aside a large part of the image space which is covered by the generative model.

3 Methods

Building upon our previous study [7], we propose a minimal predictive coding model of the visual hierarchy to account for memory replay in the visual cortex.

3.1 Model

Our model is a PCN based on a hierarchical generative model [8] with $L = 2$ layers

$$\begin{aligned} p_{\theta_0}(\mathbf{h}_0 | \mathbf{h}_1) &= \mathcal{N}(\mathbf{h}_0; \mathbf{W}_0 \mathbf{h}_1, \boldsymbol{\Sigma}_0) \\ p_{\theta_1}(\mathbf{h}_1 | \mathbf{h}_2) &= \mathcal{N}(\mathbf{h}_1; f(\mathbf{W}_1 \mathbf{h}_2 + \mathbf{b}_1), \boldsymbol{\Sigma}_1) \\ p(\mathbf{h}_2) &= \mathcal{N}(\mathbf{h}_2; \mathbf{0}, \boldsymbol{\Sigma}_2) \end{aligned}$$

where $\mathbf{h}_0, \mathbf{h}_1, \mathbf{h}_2$ are the states, $\theta_0 = \mathbf{W}_0$ and $\theta_1 = (\mathbf{W}_1, \mathbf{b}_1)$ are the parameters and f is a nonlinearity. The bias and nonlinearity are not required in level 0, which is the input level, because an image can be represented as a linear combination of basis functions [16]. The covariances $\boldsymbol{\Sigma}_i$ of the likelihoods at each level i are set to $\mathbf{I}$, except for the input level 0 during replay, as explained in section 3.3. Therefore, the prior on the latent state of level 2 is a centered isotropic multivariate Gaussian with no parameter. Recognition is assumed to be deterministic, such that

$$q_\phi(\mathbf{h}_1, \mathbf{h}_2 | \mathbf{x}) = \delta(\mathbf{h}_1 - \phi_1) \delta(\mathbf{h}_2 - \phi_2)$$

where $\phi = (\phi_1, \phi_2)$ is an estimate of the causes of the input $\mathbf{x}$.

The variational free energy for an input $\mathbf{x}$ is

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = \frac{1}{2} \boldsymbol{\xi}_0^\top \boldsymbol{\xi}_0 + \frac{1}{2} \boldsymbol{\xi}_1^\top \boldsymbol{\xi}_1 + \frac{1}{2} \boldsymbol{\xi}_2^\top \boldsymbol{\xi}_2 + \text{constant}$$

where we have defined

$$\boldsymbol{\xi}_0 = \mathbf{x} - \mathbf{W}_0 \phi_1 \tag{1}$$

$$\boldsymbol{\xi}_1 = \phi_1 - f(\mathbf{W}_1 \phi_2 + \mathbf{b}_1) \tag{2}$$

$$\boldsymbol{\xi}_2 = \phi_2. \tag{3}$$

3.2 Training algorithm

Let us consider a dataset $\mathbf{X} = \{\mathbf{x}^{(i)}\}_{i=1}^N$ of N i.i.d. samples of a continuous variable $\mathbf{x}$. The log likelihood can be written $\log p_\theta(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}) = \sum_{i=1}^N \log p_\theta(\mathbf{x}^{(i)})$. Therefore, the variational free energy of the dataset $\mathbf{X}$ is

$$\mathcal{L}(\theta; \mathbf{X}) = \sum_{i=1}^N \mathcal{L}(\theta, \phi; \mathbf{x}^{(i)}),$$

which can be estimated based on minibatches

$$\mathcal{L}(\boldsymbol{\theta}; \mathbf{X}) \approx \mathcal{L}^M(\boldsymbol{\theta}; \mathbf{X}^M) = \frac{N}{M} \sum_{i=1}^M \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}^{(i)}; \mathbf{x}^{(i)}) \quad (4)$$

where the minibatch $\mathbf{X}^M = \{\mathbf{x}^{(i)}\}_{i=1}^M$ of size M is randomly drawn from the dataset $\mathbf{X}$.

Given a minibatch $\mathbf{X}^M$, the latent states $\boldsymbol{\phi}_0^{(i)}$ and $\boldsymbol{\phi}_1^{(i)}$ are updated during T iterations for each datapoint $\mathbf{x}^{(i)}$ to minimize the variational free energy $\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}^{(i)}; \mathbf{x}^{(i)})$. For a vector $\mathbf{v}$, we note $\text{diag}[\mathbf{v}]$ the diagonal matrix with $\mathbf{v}$ in its diagonal. The update rules for the two layers in our model are

$$\Delta\boldsymbol{\phi}_1 = \alpha(\mathbf{W}_0^\top \boldsymbol{\xi}_0 - \boldsymbol{\xi}_1) \quad (5)$$

$$\Delta\boldsymbol{\phi}_2 = \alpha(\mathbf{W}_1^\top \text{diag}[f'(\mathbf{W}_1\boldsymbol{\phi}_2 + \mathbf{b}_1)] \boldsymbol{\xi}_1 - \boldsymbol{\xi}_2). \quad (6)$$

where α is the inference rate.

Then, the parameters $\boldsymbol{\theta}_0 = \mathbf{W}_0$ and $\boldsymbol{\theta}_1 = (\mathbf{W}_1, \mathbf{b}_1)$ are updated once to minimize the estimate $\mathcal{L}^M(\boldsymbol{\theta}; \mathbf{X}^M)$ given in equation 4. Thus, the learning rules can be calculated from the sum of gradients $\sum_{i=1}^M \nabla_{\boldsymbol{\theta}_l} \mathcal{L}(\boldsymbol{\theta}; \mathbf{x}^{(i)})$

$$\begin{aligned} \Delta\mathbf{W}_0 &= \beta \sum_{i=1}^M \boldsymbol{\xi}_0^{(i)} \boldsymbol{\phi}_1^{(i)\top} \\ \Delta\mathbf{W}_1 &= \beta \sum_{i=1}^M \left[\boldsymbol{\xi}_1^{(i)} \odot f'(\mathbf{W}_1\boldsymbol{\phi}_2^{(i)} + \mathbf{b}_1) \right] \boldsymbol{\phi}_2^{(i)\top} \\ \Delta\mathbf{b}_1 &= \beta \sum_{i=1}^M \boldsymbol{\xi}_1^{(i)} \odot f'(\mathbf{W}_1\boldsymbol{\phi}_2^{(i)} + \mathbf{b}_1). \end{aligned}$$

where $\odot$ is the element-wise product and β is the learning rate.

We train the model on a dataset $\mathbf{X}$ for multiple epochs, and reduce the learning rate by a multiplicative factor $\gamma < 1$ at each epoch of training to prevent instability issues. Details about the training algorithm can be found in section A.1.

3.3 Mapping to the brain

Fig. 2 shows how this algorithm can be implemented in a neural network mapped onto the visual pathway of the brain, with only local computations. The three levels in the network correspond, from bottom to top, to the lateral geniculate nucleus (LGN) in thalamus (and not the retina because it doesn't receive feedback connections from LGN), the visual cortex (VC) and the entorhinal cortex (EC). This mapping allows us to study memory replay after training the model with algorithm 1.

At inference time, the latent states are updated until convergence or until a maximum number of iterations $T_{\max}$ is reached in order to study the converged

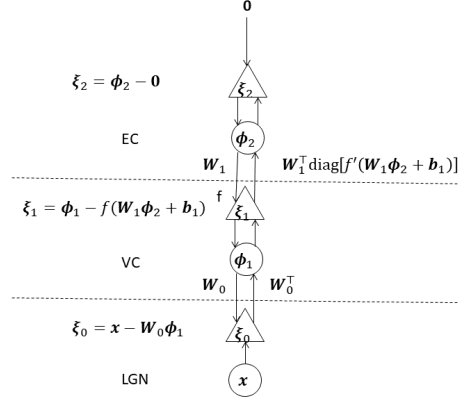


Fig. 2: PCN with 3 levels mapped to lateral geniculate nucleus (LGN), visual cortex (VC) and entorhinal cortex (EC). Each node corresponds to multiple neurons. Circle nodes correspond to units ϕ_l and triangle nodes correspond to error units ξ_l . Each connection between nodes corresponds to a fully connected network, with excitatory connections in the upward direction and inhibitory connections in the downward direction. We consider the upmost prediction to be 0, to have a standard normal distribution as prior.

representations. Convergence occurs when the relative change in the norm of the latent state ϕ_l is smaller than a threshold ϵ , i.e.

$$\frac{\|\nabla_{\phi_l} \mathcal{L}(\theta, \phi; \mathbf{x})\|}{\|\phi_l\|} < \epsilon.$$

During perception, the network is driven by the input image in LGN. While the LGN is set to the image, the VC and EC converge to hierarchical predictive representations of the image following the inference rules 5 and 6 respectively. The representation in VC is predictive of the image, as the prediction $\mathbf{W}_0 \phi_1$ is a reconstruction of the image. The representations in EC can be stored in a memory buffer representing the hippocampus and later replayed during experience replay.

Hippocampal replay is modelled as the retrieval of a pattern from the hippocampal memory buffer in the case of experience replay and as a sampling process in the case of generative replay. Thus, during memory replay, hippocampal replay first outputs the EC representation of an image retrieved from the memory buffer in the case of experience replay or a sample generated in the latent space of EC in the case of generative replay (as described in the next paragraph). Then, during cortical replay, the network is driven by the representation in EC obtained from hippocampal replay. While setting the EC level to the corresponding representation, the VC converges to the replayed representation following the inference rule 5. As the network can also be presented with an input in LGN during memory replay, the representation in VC is protected from ascending input in LGN by setting the inverse covariance, also called precision,

Σ_0^{-1} in the LGN to 0, preventing the prediction errors in LGN to influence the activity in VC. In this way, attention is focused on the representation in EC, and not on the current input. Then, the prediction $\mathbf{W}_0\phi_1$ based on the replayed representation ϕ_1 in VC corresponds to the replayed image.

During generative replay, sampling of the latent space of EC is conditioned on the class. Indeed, we fit a multivariate Gaussian distribution to each class in the latent space of EC, by estimating the mean and covariance of training samples belonging to each class. Then, samples from a given class can be generated by sampling the corresponding Gaussian distribution.

During perception, the network can also perform pattern completion when presented with a corrupted (noisy or incomplete) input. Following Salvatori et al. [19], while the LGN level is initialized to the corrupted input, the VC and EC converge following the same dynamics as in regular perception (equations 5 and 6), whereas the corrupted part of LGN converges following the inference rule

$$\Delta\phi_0 = -\alpha\phi_0.$$

obtained by calculating the gradient of the variational free energy of the input with respect to the corrupted part of LGN. Then, the prediction $\mathbf{W}_0\phi_1$ based on the recalled representation ϕ_1 in VC corresponds to the recalled image.

3.4 Experiments

The model is trained with 3 random seeds on the MNIST dataset, which contains images of handwritten digits from 0 to 9. We normalize the images before training. The original training set of 60,000 images is split into a training set of size 50,000 and a validation set of size 10,000. The validation set is used to evaluate the model during training and hyperparameter tuning. After training, we evaluate the model on the original test set of 10,000 images.

Hyperparameter values are chosen based on empirical trials informed by the predictive coding literature and summarized in Table 1. In addition, the number of hidden units in level 1 is obtained by minimizing the variational free energy on the validation set for a PCN with $L = 1$ using grid search (Fig. 6).

The choice of the number of hidden units in level 2 is more complex and results from a trade-off between different metrics. In our simulations, we study the influence of the number of hidden units in level 2 on the learned representations and on memory replay, both quantitatively and qualitatively. Quantitatively, we evaluate the predictive performance of the model, the quality of the experience replays and the pattern completion performance using the reconstruction, replay and completion errors, based on the mean squared error (MSE). The MSE between two flattened images $\mathbf{x}$ and $\hat{\mathbf{x}}$ is

$$\text{MSE}(\mathbf{x}, \hat{\mathbf{x}}) = \frac{1}{N_{\text{pixels}}} \sum_{i=1}^{N_{\text{pixels}}} (x_i - \hat{x}_i)^2.$$

The reconstruction and completion errors are computed between the original and reconstructed images and between the original and recalled images respectively, averaged over the validation set, whereas the replay error is computed

between the original and replayed images, averaged over the training set. In addition, we evaluate the linear separability of the latent manifolds corresponding to the different classes in the two hidden levels of the model using the classification accuracy of a simple multinomial logistic regression. For comparison, these measures are also computed for a one-layer PCN with the same width in level 1.

On the qualitative side, we examine examples of reconstructions of images from the validation set, as well as examples of experience replay corresponding to images from the training set. Additionally, we show examples of images generated by generative replay and visualize their hierarchical representations. Whereas the quantitative results are shown for the 3 random seeds, the qualitative results are shown for only one of the seeds.

4 Results

The reconstruction error, completion error, replay error and classification accuracy are plotted against the number of hidden units in level 2 for 3 random seeds in Fig. 3. These measures are evaluated after training the models until convergence of all layers (Fig. 7). We also include as a baseline the average values of the same metrics for a one-layer PCN with the same number of hidden units in level 1, except for the replay error because our implementation of replay requires at least two layers (see section 3.3).

On the one hand, the second hidden layer in our model does not improve the predictive power of its representations (Fig. 3, left). Indeed, the same model with the top layer removed has a lower reconstruction error than any of the models with two hidden layers. Contrary to level 1 of the model (Fig. 6), the reconstruction error on the validation set increases before decreasing as the width of level 2 is increased. In addition, the completion error increases with the number of units in the second hidden level, but it is always lower than for a model with only one hidden layer, suggesting that adding a second layer is still beneficial to the completion performance, contrarily to the reconstruction performance.

On the other hand, the quality of replay and the linear separability of the classes in the second hidden level increase with the number of units in that level, even though the linear separability remains below that of a one-layer model (Fig. 3, right). In addition, we observe that if the top level is sufficiently large, the linear separability of the classes in this level surpasses that of the level below.

In sum, the width of the top level improves the quality of replay and the linear separability of the classes in that level, but degrades the completion performance on the validation set. Therefore, choosing the number of units in level 2 based on these metrics is not straightforward, and we will turn to the qualitative evaluation.

In Fig. 4a, we visualize examples of experience replay obtained with models of different widths. As shown by Fontaine and Alexandre [7], setting the width of the top level to the number of classes (i.e. $n_2 = 10$) results in replayed images that are blurry and that do not retain the details of the original images. This issue is solved by increasing the width of the top level to 30. Increasing it further

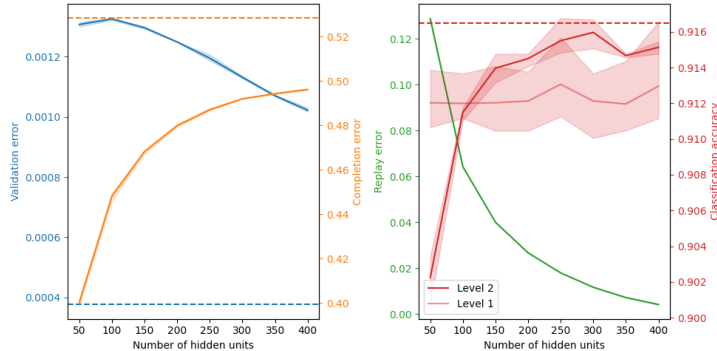


Fig. 3: Quantitative evaluation of our model according to different metrics, depending on the number of hidden units in level 2. The lines represent the averages over 3 random seeds, while the shaded areas represent the confidence interval. Dotted lines represent the average values for the same model, but with level 2 removed.

to 100 improves the sharpness of the replayed images, but at the cost of the completion performance. Indeed, while a width of 30 enables the network to semantically complete the bottom half of images taken from the validation set that were masked, a higher width of 100 makes the completion uninformative of the classes of the masked digits (Fig. 4b). Thus, the trade-off between the replay fidelity and completion performance could lead us to choose a width of 30.

Examples of reconstructions of images from the validation set and replayed images obtained by generative replay are shown for a model with $n_2 = 30$ units in level 2 (Fig. 5). Fig. 5a shows that the representations learned by the model are perfectly predictive of images it has never seen during training. In Fig. 5c, we plotted hierarchical representations obtained with generative replay, over hierarchical representations inferred from images of the validation set. The representations of the different classes are well separated in all three levels, including the top level which we previously found to have overlapping clusters due to an insufficient width [7], and the representations generated by generative replay mostly fall within the right clusters in all levels. Consequently, most of the images generated for each class are realistic examples of their classes, even though some of them are blurry (Fig. 5b).

5 Discussion

We proposed a model of hippocampo-neocortical interactions involved in memory replay using a PCN. Our work shows that predictive coding accounts for cortical replay, i.e. the transfer of an episodic memory from the hippocampus to the neocortex. To this purpose, we modelled the two modes of hippocampal replay, experience and generative replay, in a minimal way, by mapping the top level of the PCN onto the output of the hippocampus, the entorhinal cortex.

However, some of the images generated by our model were found to be out-of-distribution, probably because the simple Gaussian distribution we used does not capture the complex, non-linear geometry of the latent manifolds. This issue could be solved using other latent density models or using a Riemannian metric [1]. However, additional experiments with Fashion-MNIST and CIFAR-10 show that the complexity of the latent manifolds is increased with higher data complexity, making image generation more challenging (Fig. 8 and 9). Indeed, class separability decreases across all layers when the dataset becomes more complex (compare Fig. 3 to Fig. 10). However, for a given dataset, this separability tends to increase as we move up the network’s hierarchy and can surpass that of a one-layer PCN, especially for more complex datasets such as CIFAR-10 (Fig. 10). Therefore, our model could be adapted to more complex datasets by increasing the network depth. Nonetheless, in order to provide a more complete account of memory replay, future work should aim at modelling the hippocampal formation with its different components to understand hippocampal replay at a more detailed level.

This work also contributes to understanding the effect of depth and width on the representations learned by PCNs. Regarding the width, we showed that a larger level-2 width improves the replay quality and the linear separability of the classes. Indeed, expanding the dimensionality of the representations is known to increase their separability [4], which could cause the replay of memorized patterns to improve due to the decreased interference between them. Regarding the depth, we observed that adding a second layer does not help in making the representations of the MNIST classes more linearly separable. This could be due to the simplicity of the dataset, which is already separable at the input level (Fig. 5c). Indeed, additional experiments show that the second layer improves the linear separability of the classes only for the most complex dataset, CIFAR-10.

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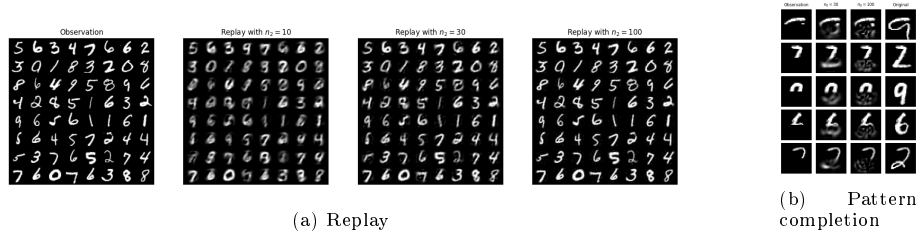


Fig.4: Replay and pattern completion by a PCN with different top level widths n_2 . (a) A minibatch of images from the training set is replayed. (b) Five images from the validation set are masked and completed by the models.

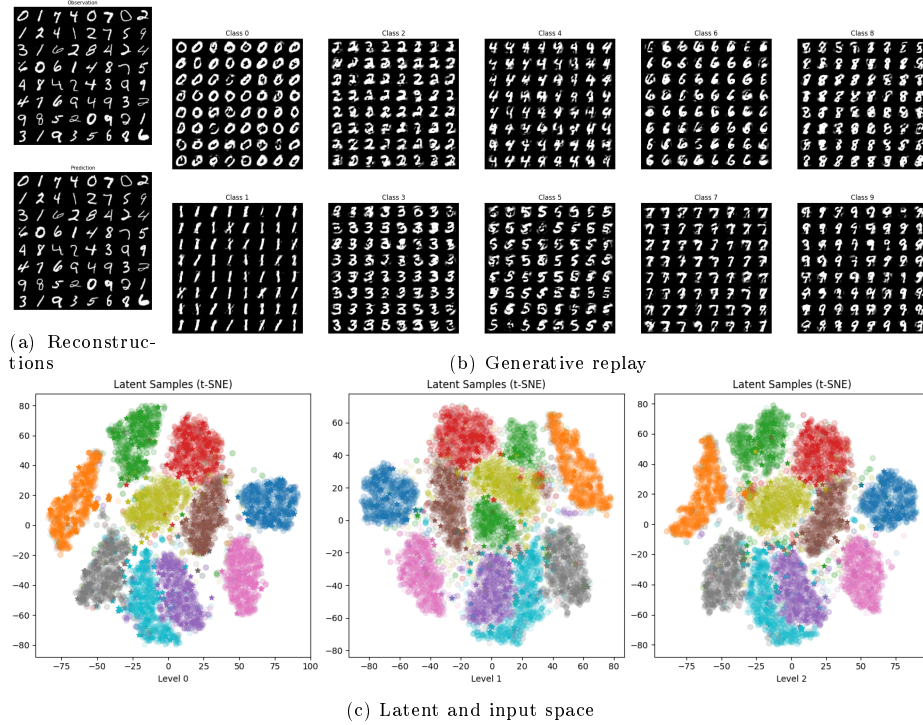


Fig.5: Visualizations for a model with $n_2 = 30$ units in the top level. (a) Reconstructions for a random mini-batch from the validation set. (b) Images generated by replay for each class. (b) Hierarchical representations of the images generated by replay (star-shaped markers) and of the images of the validation set (transparent circle markers), visualized in 2D using t-SNE. Each image is represented as one data point in each of the three subplots, colored according to the class.

A Appendix

A.1 Algorithm

The algorithm described in section 3.2 is summarized in algorithm 1 where α, β are the inference and learning rates. The constant $\frac{N}{M}$ from equation 4 is factorized in the learning rate β . To prevent instabilities which occurred systematically during training, we propose an exponential learning rate scheduler

$$\beta_{\text{epoch}} = \gamma \times \beta_{\text{epoch}-1}$$

which decays the learning rate β by a multiplicative factor γ at each epoch.

Initialization parameters include the standard deviations σ_W and σ_ϕ and the number of dimensions of latent state ϕ_0 .

Algorithm 1 Model training with minibatches

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 $W_1, W_2 \leftarrow$  Sample from  $\mathcal{N}(0, \sigma_W)$ 
 $b_1 \leftarrow$   $\mathcal{U}(-\frac{1}{n_0}, \frac{1}{n_0})$ 
repeat
  for  $k = 1$  to  $n_{\text{batches}}$  do
     $X^M \leftarrow$  Random minibatch of  $M$  datapoints drawn from  $X$ 
    for  $i = 1$  to  $M$  do
       $\phi_2^{(i)} \leftarrow x^{(i)}$ 
       $\phi_0^{(i)}, \phi_1^{(i)} \leftarrow$  Sample from  $\mathcal{N}(0, \sigma_\phi)$ 
       $\xi_0^{(i)}, \xi_1^{(i)}, \xi_2^{(i)} \leftarrow$  Calculate the corresponding errors (equation 1)
      for  $t = 1$  to  $T$  do
         $\phi_1^{(i)} \leftarrow \phi_1^{(i)} + \alpha(W_2^\top \xi_2^{(i)} - \xi_1^{(i)})$ 
         $\phi_0^{(i)} \leftarrow \phi_0^{(i)} + \alpha(W_1^\top \text{diag}[f'(W_1 \phi_0^{(i)} + b_1)] \xi_1^{(i)} - \xi_0^{(i)})$ 
         $\xi_0^{(i)}, \xi_1^{(i)}, \xi_2^{(i)} \leftarrow$  Calculate the corresponding errors (equation 1)
      end for
    end for
     $W_2 \leftarrow W_2 + \beta \sum_{i=1}^M \xi_2^{(i)} \phi_1^{(i)\top}$ 
     $W_1 \leftarrow W_1 + \beta \sum_{i=1}^M [\xi_1^{(i)} \odot f'(W_1 \phi_0^{(i)} + b_1)] \phi_0^{(i)\top}$ 
     $b_1 \leftarrow b_1 + \beta \sum_{i=1}^M \xi_1^{(i)} \odot f'(W_1 \phi_0^{(i)} + b_1)$ 
  end for
until variational free energy  $\mathcal{L}(\theta; X)$  is minimized

```

A.2 Experiments

Table 1: Hyperparameter values

Parameter	Value
Activation function f	tanh
Batch size	64
Standard deviation σ_W	0.01
Standard deviation σ_ϕ	0.05
Number of iterations T_{train}	50
Number of iterations T_{valid}	200
Maximum number of iterations T_{max}	20000
Convergence threshold ϵ	2×10^{-4}
Inference rate α	0.01
Inference optimizer	SGD
Initial learning rate β_0	10^{-5}
Learning rate decay factor γ	0.99
Learning optimizer	Adam

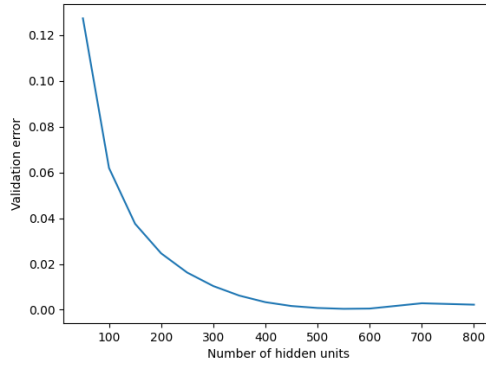


Fig. 6: RMSE between original and reconstructed images averaged over the validation set for a PCN with $L = 1$.

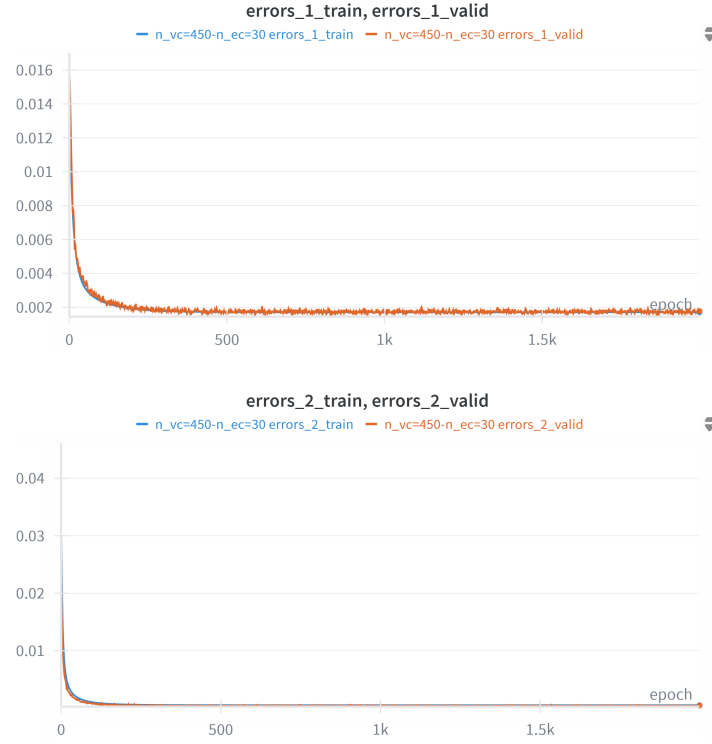


Fig. 7: Mean prediction errors in level 1 and 2 of a model with 30 units in level 0, averaged over the whole training set (blue line) and a random minibatch from the validation set (orange line) at each epoch. The model converges after 2000 epochs thanks to the learning rate scheduler.

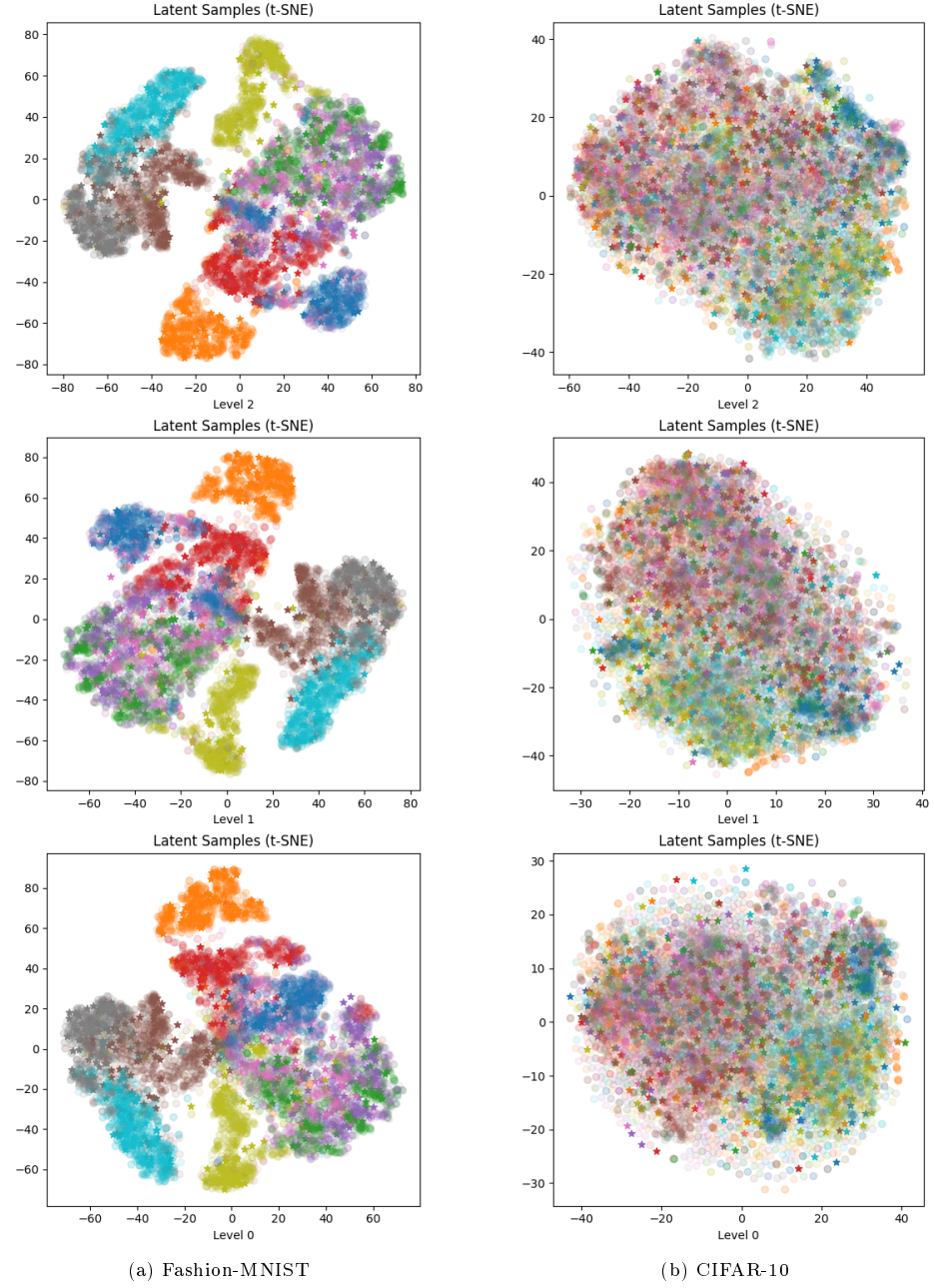


Fig. 8: Hierarchical representations of a two-layer PCN with 300 neurons in the top level, trained on (a) Fashion-MNIST and (b) CIFAR-10. See caption in Fig. 5c.

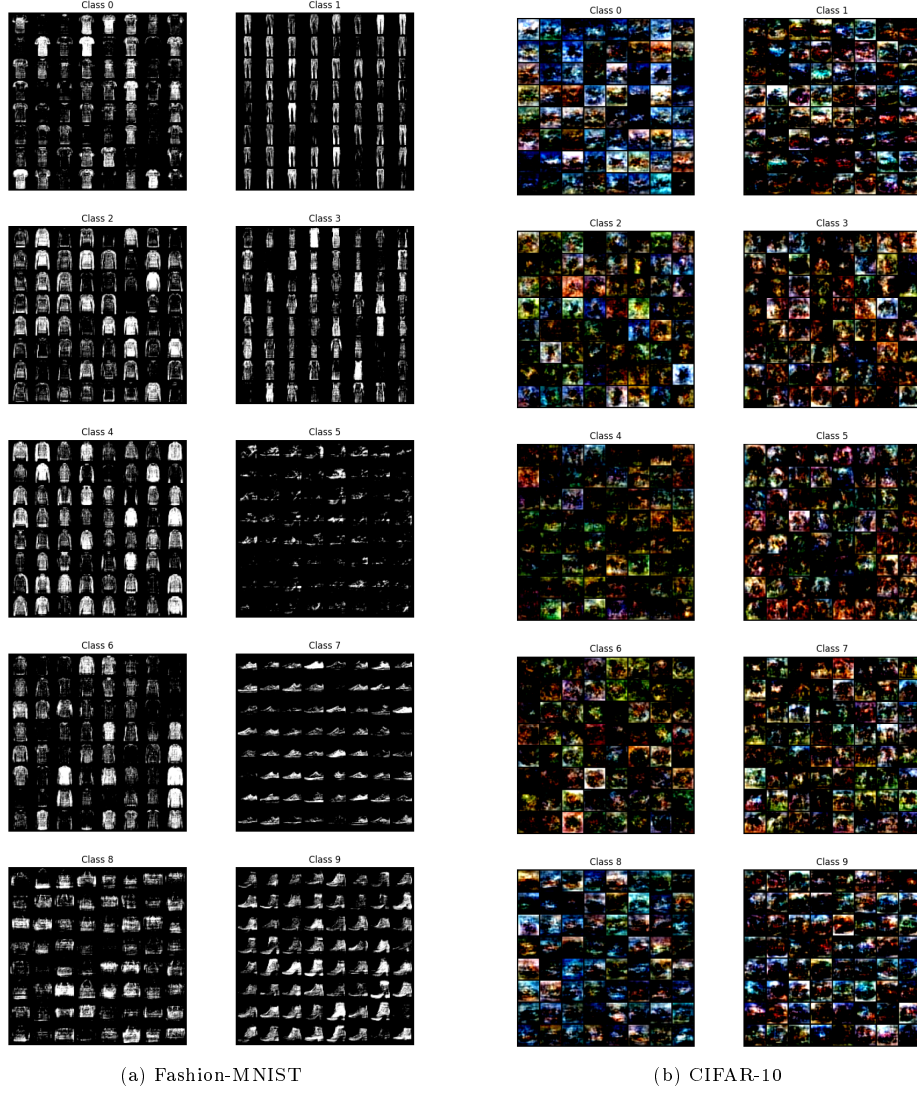


Fig. 9: Images generated by replay using a two-layer PCN with 300 neurons in the top level, trained on (a) Fashion-MNIST and (b) CIFAR-10.

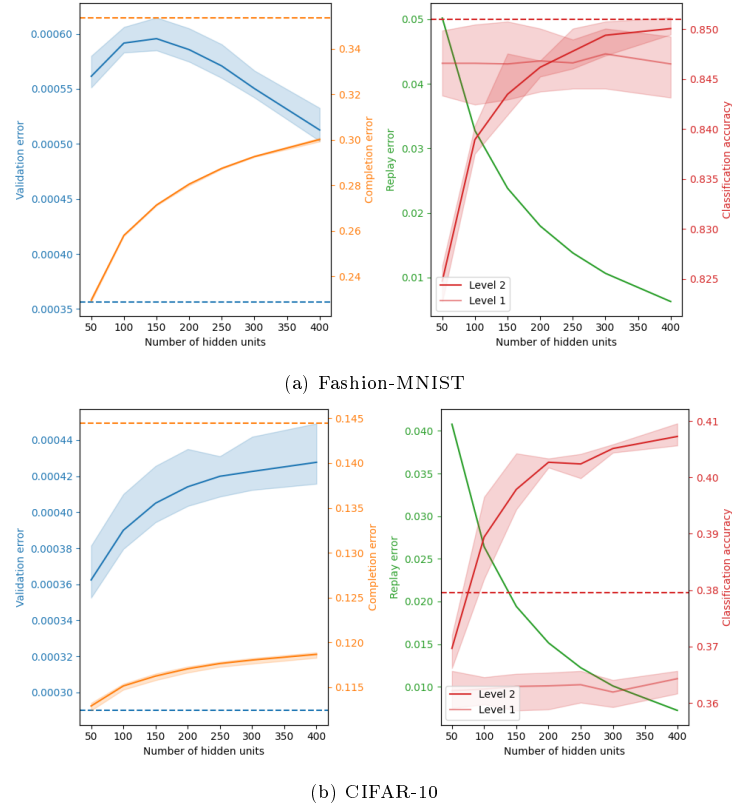


Fig. 10: Quantitative evaluation of our model trained on (a) Fashion-MNIST and (b) CIFAR-10. See caption in Fig. 3.